

Lecture 11 Collateral Constraint and Credit Cycles

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Modigliani-Miller (MM, 1958) Theorem: With a perfect and complete financial market, financial structure is irrelevant to real outcomes.

Perfect substitutability across fundings.
Cost of capital is the same. ... no-arbitrage condition of holding capital and bond.

Asymmetric information or imperfect enforceability of financial contracts break MM theorem.

Kiyotaki-Moore (1997, JPE) financial market imperfection. ... amplification & persistence of shocks

Capital: ① production factor ② Collateral for loans

Negative productivity shock: output $\downarrow \Rightarrow$ value of collateral $\downarrow \Rightarrow$ borrowing $\downarrow \Rightarrow$ output $\downarrow$
financial multiplier that amplifies shock.

Measure 1 of farmers: $E_t \sum_{s=0}^{\infty} \beta^s X_{t+s}$

Measure n of gatherers: $E_t \sum_{s=0}^{\infty} \beta'^s X'_{t+s}$

Farmers are more impatient $\beta < \beta'$

$\Rightarrow$ farmers are borrowers.

Both use land k_t to produce fruit, value of land $q_t k_t$ as collateral

Production takes one period: $y_{t+1} = (a+c)k_t$

$a k_t$ = sellable fruit $c k_t$ = bruised fruit must be consumed.

Assume $c > a(\frac{1}{\beta} - 1)$ farmer wants to consume $c k_t$ and sell $a k_t$

Constraint farmer: $(a+c)k_{t+1} + b_t + q_t k_{t+1} = X_t + R b_{t-1} + q_t k_t$... budget constraint.

$R b_t \leq q_{t+1} k_{t+1}$... borrowing constraint.

Unconstrained gatherers: $y'_{t+1} = G(c k'_t)$. $G(\cdot)$ decreasing return to scale.

$G(c k'_{t+1}) + b'_t + q_t k'_{t+1} = X'_t + R b'_{t-1} + q_t k'_t$

Equilibrium definition: Equilibrium consists of allocations of land, debt, consumption $\{q_t, k_t, k'_t, b_t, b'_t, X_t, X'_t\}$ such that:

1. Given q_t , farmers choose $\{k_t, b_t, X_t\}$ to maximize utility subject to the production function, borrowing constraint, budget constraint.

2. Given q_t , gatherers choose $\{k'_t, b'_t, X'_t\}$ to maximize utility subject to production function and budget constraint

3. Market for land, fruit and debt clears

4. Perfect foresight, no uncertainty.

Equilibrium characterization:

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1° Farmers always borrow maximum and invest in land:

$$b_t = \frac{q_{t+1}k_t}{R} \quad \text{and} \quad x_t = c k_{t-1}$$

From farmer's BC: $a k_{t-1} + \frac{q_{t+1}k_t}{R} + q_t k_{t-1} = R \frac{q_t k_{t-1}}{R} + q_t k_t$

$$\Rightarrow k_t = \frac{1}{q_t - \frac{q_{t+1}}{R}} \underbrace{[(a + q_t) k_{t-1} - R b_{t-1}]}_{\text{Net Worth.}}$$

new land price.

$$u_t = q_t - \frac{q_{t+1}}{R} \dots \text{down payment}$$

amount that can borrow against with new land.

2° Gatherers' demand for land:

$$\frac{G'(c k_{t+1}')}{R} = u_t = \underbrace{q_t}_{\text{buying cost}} - \underbrace{\frac{q_{t+1}}{R}}_{\text{user cost}}$$

extra output of fruit $\rightarrow$ resell value of land

$$\mathcal{L} = \sum_{s=0}^{\infty} \beta^s \left\{ x_{t+s}' + \lambda_{t+s} [G(c k_{t+s}') + b_t' + q_t k_{t-1}' - x_t' - R b_{t-1}' - q_t k_t'] \right\}$$

F.O.C.

$$\frac{\partial \mathcal{L}}{\partial k_t'} = 0 \Rightarrow \lambda_{t+1} G'(c k_{t+1}') + \lambda_{t+1} q_{t+1} - \lambda_t q_t = 0.$$

$$\frac{\partial \mathcal{L}}{\partial b_t} = 0 \Rightarrow \lambda_t - \lambda_{t+1} R = 0 \Rightarrow \frac{\lambda_{t+1}}{\lambda_t} = \frac{1}{R}$$

$$\Rightarrow \frac{G'(c k_{t+1}')}{R} + \frac{q_{t+1}}{R} = q_t \Rightarrow \frac{G'(c k_{t+1}')}{R} = q_t - \frac{q_{t+1}}{R}$$

* One unit of internal funds buy $\frac{1}{q_t - \frac{q_{t+1}}{R}}$ units of land

Give up 1 consumption and buy land, next period yield: $\frac{a+c}{u_t} \Rightarrow$ discounted $\beta \frac{a+c}{u_t} > 1$
means investing rather than consuming.

Farmers in the aggregate:

$$k_t = \frac{1}{u_t} [(a + q_t) k_{t-1} - R b_{t-1}]$$

$$b_t = \frac{1}{R} q_{t+1} k_t$$

Land market resource constraint: $m k_t' + k_t = \bar{K}$

Land market clearing: $u_t = q_t - \frac{q_{t+1}}{R} = \frac{G'(\frac{1}{m}(\bar{K} - k_t))}{R}$

No bubbles in land price: $\lim_{s \rightarrow \infty} E_t (R^{-s} q_{t+s}) = 0$

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$$\Rightarrow \hat{q}_t + \hat{k}_t = \frac{a\Delta + q_t - q^*}{u^* k^*} k^* = \frac{a\Delta + q_t - q^*}{a k^*} k^* = \Delta + \frac{q_t - q^*}{a}$$

$$= \Delta + \frac{q^*}{a} \cdot \hat{q}_t = \Delta + \frac{R}{R-1} \hat{q}_t \quad \text{Let } \eta = \frac{dk_t}{du_t} \frac{u^*}{k^*} \quad \text{Elasticity of land supply of gatherer w.r.t. user cost}$$

$$\Rightarrow \hat{k}_t = \eta \cdot \hat{u}_t \text{ around S.S.} \Rightarrow \hat{u}_t = \frac{1}{\eta} \hat{k}_t \text{ plug in}$$

$\Rightarrow (1 + \frac{1}{\eta}) \hat{k}_t = \Delta + \frac{R}{R-1} \hat{q}_t$ $\eta \uparrow$ more elastic land supply from gatherers.
 a small increase in user cost induces big release of land.
 Higher k_t = direct productivity effect + collateral / asset price effect (higher land price $\Rightarrow$ higher collateral value)
 A modest R can lead to huge amplification; $R = 1.05 \Rightarrow \frac{R}{R-1} = 21$

Log linearize ②: $u(k_{t+s})k_{t+s} = a k_{t+s-1} \Rightarrow u^* k^* (\hat{u}_{t+s} + \hat{k}_{t+s}) = a k^* \hat{k}_{t+s-1} \Rightarrow \hat{k}_{t+s} = \frac{\eta}{1+\eta} \hat{k}_{t+s-1}$
 Original shock Δ disappears, land holding decay gradually, $\frac{\eta}{1+\eta} \in (0, 1)$ $\eta \uparrow \Rightarrow$ decay slow $\Rightarrow$ persistence effect

Land price respond in the same way as user's cost: $\hat{q}_t = \frac{1}{\eta} \Delta$

$$\Rightarrow \hat{k}_t = \frac{\eta}{1+\eta} (1 + \frac{R}{R-1} \frac{1}{\eta}) \Delta \quad \text{when } \eta = 1 \quad R = 1.05 \Rightarrow \hat{k}_t \approx 11 \Delta$$

Aggregate output: $Y_{t+s} = (a+c)k_{t+s-1} + m G(k_{t+s-1}) = (a+c)k_{t+s-1} + m \cdot G(\frac{\bar{k} - k_{t+s-1}}{m})$

$$\frac{dY}{dk} \Big|_{k^*} = (a+c) - G'(\frac{\bar{k} - k^*}{m}) \quad \text{since } u^* = \frac{G'(\frac{\bar{k} - k^*}{m})}{R} = a \Rightarrow Y_{t+s} - Y^* = [(a+c) - Ra](k_{t+s-1} - k^*)$$

$$\Rightarrow \hat{Y}_{t+s} = \frac{(a+c) - Ra}{Y^*} (k_{t+s-1} - k^*) = \frac{(a+c) - Ra}{Y^*} k^* \hat{k}_{t+s-1} = \frac{a+c-Ra}{a+c} \frac{(a+c)k^*}{Y^*} \hat{k}_{t+s-1}$$

$a+c-Ra$: net output gain in reallocating Land from gatherers to farmers

$\frac{(a+c)k^*}{Y^*}$: scaling term, how important is farmer's production in total S.S. output

Generates output persistence from land reallocation

b) Static response of land price and land holding. Suppose $q_{t+1} = q^*$

From: $u(k_t)k_t = (a + \Delta a + q_t - q^*)k^* \Rightarrow (1 + \frac{1}{\eta}) \hat{k}_t = \Delta + \frac{R}{R-1} (\frac{R-1}{R} \frac{1}{\eta} \hat{k}_t)$

$$\left. \begin{array}{l} u_t = q_t - \frac{q^*}{R} \\ u_t - u^* = q_t - q^* \end{array} \right\} \Rightarrow \hat{u}_t = \frac{R}{R-1} \hat{q}_t \quad \Rightarrow (1 + \frac{1}{\eta}) \hat{k}_t = \Delta + \frac{1}{\eta} \hat{k}_t \Rightarrow \hat{k}_t = \Delta$$

$$\hat{q}_t = \frac{R-1}{R} \frac{1}{\eta} \hat{k}_t \quad \Rightarrow \hat{k}_t |_{q_{t+1}=q^*} = \Delta \quad \text{One-for-one response without amplification.}$$

Plug into: $\hat{q}_t = \frac{R-1}{R} \frac{1}{\eta} \hat{k}_t \Rightarrow \hat{q}_t |_{q_{t+1}=q^*} = \frac{R-1}{R} \frac{1}{\eta} \Delta$

Compare $\hat{q}_t = \frac{1}{\eta} \Delta$

$\Rightarrow \frac{R-1}{R} \approx 0.048$ when $R = 1.05$
 $\Rightarrow$ static price response is just 4.8% of the full model response

* Land-price jump in the full model comes from the forward-looking collateral mechanism.

At first-best steady state, unexpected increase in productivity only rises output by Δ , prices q^0 and land k^0 unaffected, because both agents are constraint in the credit market.

Fit MM theorem that change in net worth does not affect real terms.